CIRC HQ STUDENT HANDOUT | GRADE 5

Cardboard Micro-Furniture

Goal: Build a miniature cardboard model of seating, storage, a work surface, or a quiet-space feature for a specific classroom user.

35 minutes | 8 steps | Engineering and Design | Playbook A meeting 8, grade 5 path

Ask | Plan | Build | Test | Improve

You are done when

Name the user, location, and barrier, then demonstrate how the revised model improves access or function.

You will use

- Cardboard
- Paper user figures
- User cards
- Rulers
- Scissors
- Tape
- Paper fasteners
- Grid paper
- Markers

Steps

35 minutes total. Your teacher says when to move on.

1 Step 1: Get ready — 3 min

- Choose one user card.
- Choose one classroom location.

2 Step 2: Cut safely — 3 min

- Cut only on the table.
- Keep fingers outside the cutting path.
- Carry scissors closed.

3 Step 3: Plan for the user — 4 min

- Draw the furniture feature.
- Label how the user reaches it.
- Mark the model size.

4 Step 4: Build the model — 9 min

- Cut the planned cardboard shapes.
- Use two connection methods.
- Keep the model table sized.

5 Step 5: Test the user path — 6 min

- Move the paper user toward the model.
- Check the reach point.
- Check the model stability.

6 Step 6: Remove a problem — 4 min

- Change one blocked or weak feature.
- Repeat the user path.

7 Step 7: Sort the materials — 3 min

- Close the scissors.
- Return usable cardboard.
- Store the model flat.

8 Step 8: Exit check — 3 min

- Point to the user feature.
- Explain the problem you changed.

What to turn in

A mini model tested with a paper user figure.

No computer? Another way to do it

Draw and label the design for the user.

Finished early? Next challenge

Second User: Add one small change that makes the feature easier for a second user with a different reach or movement need.

Stay safe

Cut only on the table, use fasteners as demonstrated, and keep models small enough to test safely at the workspace.

Stuck? Try this order

1. Read the step again, out loud if you can.
2. Ask a partner at your table.
3. Raise your hand. Keep working on the next part while you wait.